

Chapter 14

Ratio, Similarity and Symmetry

For comparing two quantities, their ratios are to be considered. Again, for determining ratios, the two quantities are to be measured in the same units. In algebra we have discussed this in detail.

At the end of this chapter, the students will be able to —

- ▶ explain geometric ratios;
- ▶ explain the internal division of a line segment;
- ▶ verify and prove theorems related to ratios;
- ▶ verify and prove theorems related to similarity;
- ▶ explain the concepts of symmetry;
- ▶ verify line and rotational symmetry of real objects practically.

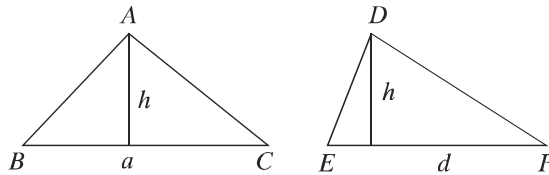
Properties of Ratio and Proportion

- (i) If $a : b = x : y$ and $c : d = x : y$, it follows that, $a : b = c : d$.
- (ii) If $a : b = b : a$, it follows that, $a = b$
- (iii) If $a : b = x : y$, it follows that, $b : a = y : x$ (inversendo).
- (iv) If $a : b = x : y$, it follows that, $a : x = b : y$ (alternendo).
- (v) If $a : b = c : d$, it follows that, $ad = bc$ (cross-multiplication)
- (vi) If $a : b = x : y$, it follows that, $a + b : b = x + y : y$ (componendo)
and $a - b : b = x - y : y$ (dividendo)
- (vii) If $\frac{a}{b} = \frac{c}{d}$, it follows that $\frac{a+b}{a-b} = \frac{c+d}{c-d}$ (componendo and dividendo).

Geometric proportions

We have learnt to find the area of a triangular region. Two necessary concepts of ratio are to be formed from this.

1. If the heights of two triangles are equal, their bases and areas are proportional.

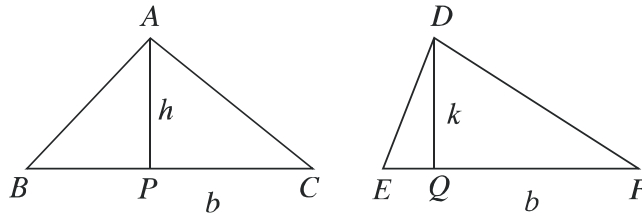


Let the bases of the triangles ABC and DEF be $BC = a$, $EF = d$ respectively and the height in both cases be h .

Hence, the area of the triangle $ABC = \frac{1}{2} \times a \times h$, the area of the triangle $DEF = \frac{1}{2} \times d \times h$

Therefore, the area of the triangle ABC : area of the triangle DEF
 $= \frac{1}{2} \times a \times h : \frac{1}{2} \times d \times h = a : d = BC : EF$

2. If the bases of two triangles are equal, their heights and areas are proportional.



Let the heights of the triangles ABC and DEF be $AP = h$, $DQ = k$ respectively and the base in both cases be b .

Hence, the area of the triangle $ABC = \frac{1}{2} \times b \times h$, and the area of the triangle $DEF = \frac{1}{2} \times b \times k$

Therefore, the area of the triangle ABC : area of the triangle DEF
 $= \frac{1}{2} \times b \times h : \frac{1}{2} \times b \times k = h : k = AP : DQ$

Theorem 28. A straight line drawn parallel to one side of a triangle intersects the other two sides or those sides produced proportionally.

Special Nomination: In the figure, the straight line DE is parallel to the side BC of the triangle ABC . DE intersects AB and AC (figure-1) or their produced sections (figure-2) at D and E respectively. It is required to prove that $AD : DB = AE : EC$

Drawing: Join B, E and C, D .

Proof:

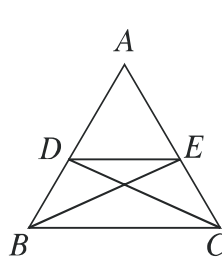


Figure 1

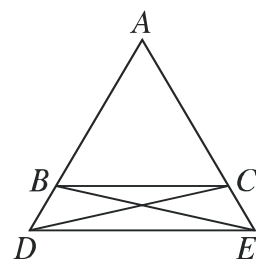


Figure 2

Step 1. The heights of $\triangle ADE$ and $\triangle BDE$ are equal.

$$\therefore \frac{\triangle ADE}{\triangle BDE} = \frac{AD}{DB} \quad [\text{The bases of the triangles of equal height are proportional}]$$

Step 2. The heights of $\triangle ADE$ and $\triangle DEC$ are equal

$$\therefore \frac{\triangle ADE}{\triangle DEC} = \frac{AE}{EC} \quad [\text{The bases of the triangles of equal height are proportional}]$$

Step 3. But $\triangle BDE = \triangle DEC$ [On the same base DE and between same pair of lines]

$$\therefore \frac{\triangle ADE}{\triangle BDE} = \frac{\triangle ADE}{\triangle DEC}$$

Step 4. Therefore, $\frac{AD}{DB} = \frac{AE}{EC}$

i.e, $AD : DB = AE : EC$

Corollary 1. If the line parallel to BC of the triangle ABC intersects the sides AB and AC at D and E respectively, then $\frac{AB}{AD} = \frac{AC}{AE}$ and $\frac{AB}{BD} = \frac{AC}{CE}$.

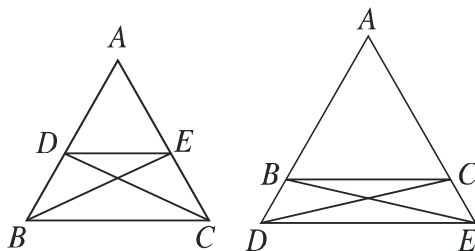
Corollary 2. The line through the midpoint of a side of a triangle parallel to another side bisects the third line.

The proposition opposite of theorem 28 is also true. That is if a line segment divides the two sides of a triangle or the line produced proportionally, it is parallel to the third side. Here follows the proof of the theorem.

Theorem 29. If a line segment divides the two sides or their produced sections of a triangle proportionally, it is parallel to the third side.

Special Nomination: In the triangle ABC the line segment DE divides the two sides AB and AC or their produced sections proportionally. That is, $AD : DB = AE : EC$. It is required to prove that DE and BC are parallel.

Drawing: Join B, E and C, D .



Proof:

Step 1. $\frac{\triangle ADE}{\triangle BDE} = \frac{AD}{DB}$ [Triangles with equal height]

and $\frac{\triangle ADE}{\triangle DEC} = \frac{AE}{EC}$ [Triangles with equal height]

Step 2. But $\frac{AD}{DB} = \frac{AE}{EC}$ [Given]

Step 3. Therefore, $\frac{\triangle ADE}{\triangle BDE} = \frac{\triangle ADE}{\triangle DEC}$ [from (i) and (ii)]

$$\therefore \triangle BDE = \triangle DEC$$

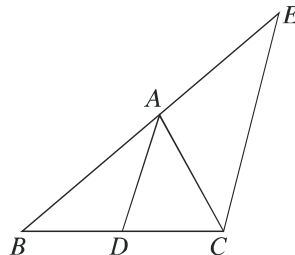
Step 4. But $\triangle BDE$ and $\triangle DEC$ are on the same side of the common base DE .
So they lie between a pair of parallel lines.

$\therefore BC$ and DE are parallel.

Theorem 30. The internal bisector of an angle of a triangle divides its opposite side in the ratio of the sides constituting to the angle.

Special Nomination: In $\triangle ABC$ the line segment AD bisects the internal angle $\angle A$ and intersects the side BC at D . It is required to prove that $BD : DC = BA : AC$.

Drawing: Draw the line segment CE parallel to DA , so that it intersects the side BA produced at E .



Proof:

Step 1. Since, $DA \parallel CE$ and BE is their transversal [by construction]

$$\therefore \angle AEC = \angle BAD \quad [\text{corresponding angles}]$$

Again, $DA \parallel CE$ and AC is their transversal $\therefore \angle ACE = \angle CAD$
[alternate angles]

Step 2. But $\angle BAD = \angle CAD$ [supposition]

$$\therefore \angle AEC = \angle ACE \quad \text{Hence, } AC = AE \quad [\text{Chapter 6 Theorem 8}]$$

Step 3. Again, since $DA \parallel CE$ Therefore, $\frac{BD}{DC} = \frac{BA}{AE}$ [step 2]

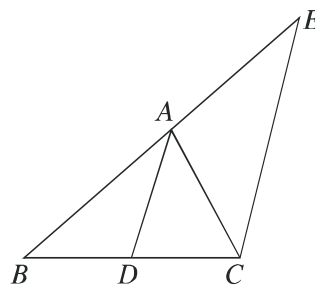
Step 4. But $AE = AC$

$$\therefore \frac{BD}{DC} = \frac{BA}{AC}$$

Theorem 31. If any side of a triangle is divided internally, in the ratio of the other two sides, the line segment from the point of division to the opposite vertex bisects the angle at the vertex.

Special Nomination: Let ABC be a triangle and the line segment AD from vertex A divides the side BC at D such that $BD : DC = BA : AC$. It is required to prove that AD bisects $\angle BAC$ i.e. $\angle BAD = \angle CAD$

Drawing: Draw at C the line segment CE parallel to DA , so that it intersects the side BA produced at E .

**Proof:**

Step 1. For $\triangle BCE$ $DA \parallel CE$ [by construction]

$$\therefore BA : AE = BD : DC \quad [\text{theorem 28}]$$

Step 2. But $BD : DC = BA : AC$ [supposition]

$$\therefore BA : AE = BA : AC \quad [\text{from step 1 and step 2}]$$

$$\therefore AE = AC$$

Therefore, $\angle ACE = \angle AEC$ [base angles of isosceles triangle are equal]

Step 3. But $\angle AEC = \angle BAD$ [corresponding angles]

and $\angle ACE = \angle CAD$ [alternate angles]

Therefore, $\angle BAD = \angle CAD$ [from step 2]

$\therefore$ the line segment AD bisects $\angle BAC$.

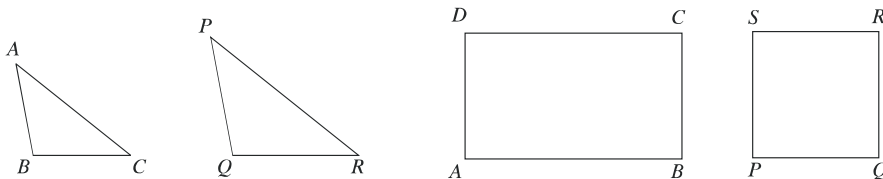
Exercise 14.1

1. The bisectors of two base angles of a triangle intersect the opposite sides at X and Y respectively. If XY is parallel to the base, prove that the triangle is an isosceles triangle.
2. Prove that if two lines intersect a few parallel lines, the matching sides are proportional.
3. Prove that the diagonals of a trapezium are divided in the same ratio at their point of intersection.
4. Prove that the line segment joining the midpoints of oblique sides of a trapezium and two parallel sides are parallel.
5. The medians AD and BE of the triangle ABC intersect each other at G . A line segment is drawn through G parallel to DE which intersects AC at F . Prove that $AC = 6EF$.
6. In the triangle ABC , X is any point on BC and O is a point on AX . Prove that $\triangle AOB : \triangle AOC = BX : XC$
7. In the triangle ABC , the bisector of A intersects BC at D . A line segment drawn parallel to BC intersects AB and AC at E and F respectively. Prove that $BD : DC = BE : CF$.

Similarity

The congruence and similarity of triangles have been discussed earlier in class VII. In general, congruence is a special case of similarity. If two figures are congruent, they are similar; but two similar figures are not always congruent.

Equiangular Polygons: If the angles of two polygons with equal number of sides are sequentially equal, the polygons are known as equiangular polygons.



Similar Polygons: If the vertices of two polygons with equal number of sides can be matched in such a sequential way that

- (i) The matching angles are equal and (ii) The ratios of matching sides are equal, then the two polygons are called similar polygons.

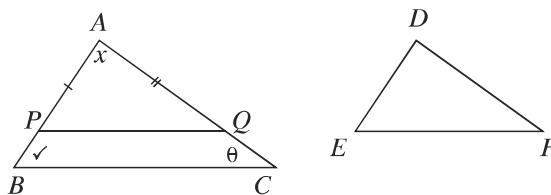
In the above figures, the rectangle $ABCD$ and the square $PQRS$ are equiangular since the number of sides in both the figures is 4 and the angles of the rectangle are sequentially equal to the angles of the square (all right angles). Though the similar angles of the figure are equal, the ratios of the matching sides are not the same. Hence the figures are not similar. In case of triangles, situation like this does not arise. As a result of matching the vertices of triangles, if one of the conditions of similarity is true, the other condition automatically becomes true and the triangles are similar. That is, two similar triangles are always equiangular and two equiangular triangles are always similar.

If two triangles are equiangular and one of their matching pairs is equal, the triangles are congruent. The ratio of the matching sides of two equiangular triangles is a constant. Proofs of the related theorems are given below.

Theorem 32. If two triangles are equiangular, their matching sides are proportional.

Special Nomination: Let ABC and DEF be triangles with $\angle A = \angle D$, $\angle B = \angle E$ and $\angle C = \angle F$

We need to prove that, $\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$



Drawing: Consider the matching sides of the triangles ABC and DEF unequal. Take two points P and Q on AB and AC respectively so that $AP = DE$ and $AQ = DF$. Join P and Q and complete the drawing.

Proof:

Step 1. In $\triangle APQ$ and $\triangle DEF$

$AP=DE, AQ=DF, \angle A = \angle D$ Therefore, $\triangle APQ \cong \triangle DEF$ [SAS theorem]

Therefore, $\angle APQ = \angle DEF = \angle ABC$ and $\angle AQP = \angle DFE = \angle ACB$.

That is, the corresponding angles produced as a result of intersections of AB and AC by the line segment PQ are equal.

Therefore $PQ \parallel BC$ $\therefore \frac{AB}{AP} = \frac{AC}{AQ}$ or, $\frac{AB}{DE} = \frac{AC}{DF}$ [corollary 1]

Step 2. Similarly, cutting line segments ED and EF from BA and BC respectively, it can be shown that,

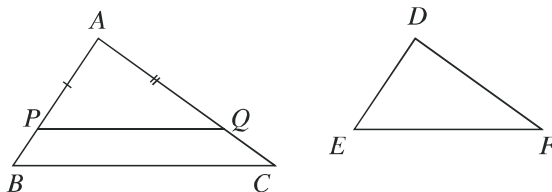
$$\frac{BA}{ED} = \frac{BC}{EF} \quad [\text{theorem 28}]$$

$$\text{i.e. } \frac{AB}{DE} = \frac{BC}{EF} \quad \therefore \frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$$

The proposition opposite of theorem 32 is also true.

Theorem 33. If the sides of two triangles are proportional, the opposite angles of their matching sides are equal.

Special Nomination: Let, in $\triangle ABC$ and $\triangle DEF$, $\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$. It is to prove that, $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$.



Drawing: Consider the matching sides of the triangles ABC and DEF unequal. Take two points P and Q on AB and AC respectively, so that $AP = DE$ and $AQ = DF$. Join P and Q to complete the drawing.

Proof:

Since $\frac{AB}{DE} = \frac{AC}{DF}$, Therefore $\frac{AB}{AP} = \frac{AC}{AQ}$

Therefore $PQ \parallel BC$ [theorem 29]

$\therefore \angle ABC = \angle APQ$ [corresponding angles made by the transversal AB]

and $\angle ACB = \angle AQP$ [corresponding angles made by the transversal AC]

$\therefore \triangle ABC$ and $\triangle APQ$ are equiangular.

Therefore, $\frac{AB}{AP} = \frac{BC}{PQ}$ or, $\frac{AB}{DE} = \frac{BC}{PQ}$ [theorem 32]

But $\frac{AB}{DE} = \frac{BC}{EF}$ [supposition]

$$\therefore \frac{BC}{EF} = \frac{BC}{PQ}$$

$$\therefore EF = PQ$$

Therefore $\triangle APQ$ and $\triangle DEF$ are congruent. [SSS theorem]

$$\therefore \angle PAQ = \angle EDF, \angle APQ = \angle DEF, \angle AQP = \angle DFE$$

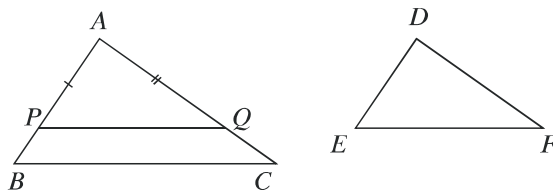
$$\therefore \angle APQ = \angle ABC \text{ and } \angle AQP = \angle ACB$$

$$\angle A = \angle D, \angle B = \angle E, \angle C = \angle F$$

Theorem 34. If one angle of a triangle is equal to an angle of the other and the sides adjacent to the equal angles are proportional, the triangles are similar.

Special Nomination: Let in $\triangle ABC$ and $\triangle DEF$, $\angle A = \angle D$ and $\frac{AB}{DE} = \frac{AC}{DF}$.

It is to be proved that, $\triangle ABC$ and $\triangle DEF$ are similar.



Drawing: Consider the matching sides of ABC and DEF unequal. Take two points P and Q on AB and AC respectively so that $AP = DE$ and $AQ = DF$. Join P and Q to complete the drawing.

Proof:

For $\triangle APQ$ and $\triangle DEF$, $AP = DE$, $AQ = DF$ and internal $\angle A = \text{internal } \angle D$

$$\therefore \triangle APQ \cong \triangle DEF \quad [\text{SAS theorem}]$$

$$\therefore \angle A = \angle D, \angle APQ = \angle E, \angle AQP = \angle F$$

Again, since $\frac{AB}{DE} = \frac{AC}{DF}$, therefore $\frac{AB}{AP} = \frac{AC}{AQ}$ [theorem 29]

$$\therefore PQ \parallel BC$$

Therefore $\angle ABC = \angle APQ$ and $\angle ACB = \angle AQP$

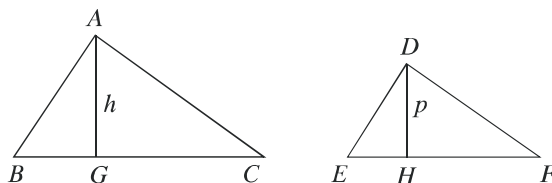
$\therefore \angle A = \angle D$, $\angle B = \angle E$, and $\angle C = \angle F$

i.e. $\triangle ABC$ and $\triangle DEF$ are equiangular.

Therefore $\triangle ABC$ and $\triangle DEF$ are similar.

Theorem 35. The ratio of the areas of two similar triangles is equal to the ratio of squares on any two matching sides.

Special Nomination: Let the triangles ABC and DEF be similar and BC and EF be their matching sides respectively. It is required to prove that, $\triangle ABC : \triangle DEF = BC^2 : EF^2$



Drawing: Draw perpendiculars AG and DH on BC and EF respectively. Let $AG = h$, $DH = p$.

Proof:

Step 1. $\triangle ABC = \frac{1}{2} \times BC \times h$ and $\triangle DEF = \frac{1}{2} \times EF \times p$

$$\frac{\triangle ABC}{\triangle DEF} = \frac{\frac{1}{2} \times BC \times h}{\frac{1}{2} \times EF \times p} = \frac{h}{p} \times \frac{BC}{EF}$$

Step 2. But in the triangles ABG and DEH , $\angle B = \angle E$, $\angle AGB = \angle DHE$
[right angle]

$\therefore \angle BAG = \angle EDH$

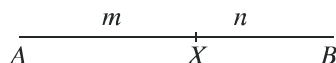
$\therefore \triangle ABC$ and $\triangle DEF$ are equiangular, so similar.

$$\therefore \frac{h}{p} = \frac{AB}{DE} = \frac{BC}{EF} \quad [\text{as } \triangle ABC \text{ and } \triangle DEF \text{ are similar}]$$

Step 3. $\frac{\triangle ABC}{\triangle DEF} = \frac{h}{p} \times \frac{BC}{EF} = \frac{BC}{EF} \times \frac{BC}{EF} = \frac{BC^2}{EF^2}$

Internal Division of a Line Segment in definite ratio

If A and B are two different points in a plane and m and n are two natural numbers, we acknowledge that there exists a unique point X lying between A and B and $AX : XB = m : n$.

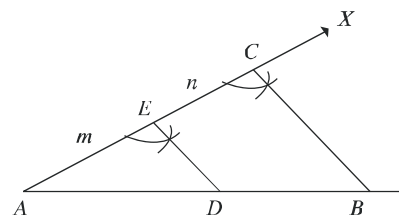


In the above figure, the line segment AB is divided at X internally in the ratio $m : n$ i.e. $AX : XB = m : n$.

Construction 12. To divide a given line segment internally in a given ratio.

Special Nomination: Let the line segment AB be divided internally in the ratio $m : n$.

Drawing: Let an angle BAX be drawn at point A . From AX cut the lengths $AE = m$ and $EC = n$ sequentially. Join B, C . At E , draw line segment ED parallel to CB which intersects AB at D . Then the line segment AB is divided at D internally in the ratio $m : n$.



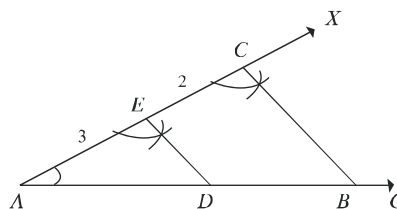
Proof: Since the line segment DE is parallel to a side BC of the triangle ABC .

$\therefore AD : DB = AE : EC = m : n$

Work: Divide a given line segment in definite ratio internally by an alternative method.

Example 1. Divide a line segment of length 7 c m internally in the ratio 3 : 2.

Solution: Draw any ray AG . From AG , cut a line segment $AB = 7$ c m. Draw an angle $\angle BAX$ at A . From AX , cut the lengths $AE = 3$ c.m. and $EC = 2$ c m from EX . Join B, C . At E , draw an $\angle AED$ equal to $\angle ACB$ whose side intersects AB at D . Then the line segment AB is divided at D internally in the ratio 3 : 2.



Work:

Draw a triangle similar to a particular triangle whose sides are $\frac{3}{5} \times$ the sides of the given triangle.

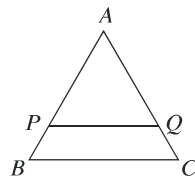
Exercise 14.2

1. In $\triangle ABC$, if the line DE parallel to BC intersects AB and AC at D and E respectively, then-

- (i) $\triangle ABC$ and $\triangle ADE$ are similar.
 (ii) $\frac{AD}{BD} = \frac{CE}{AE}$
 (iii) $\frac{\triangle ABC}{\triangle ADE} = \frac{BC^2}{DE^2}$

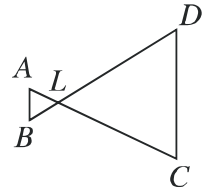
Which one of the following is true?

- 1) i and ii 2) i and iii 3) ii and iii 4) i , ii and iii
2. If in $\triangle ABC$, $PQ \parallel BC$, then which one of the following is true?
- 1) $AP : PB = AQ : QC$
 2) $AB : PQ = AC : PQ$
 3) $AB : AC = PQ : BC$
 4) $PQ : BC = BP : BQ$



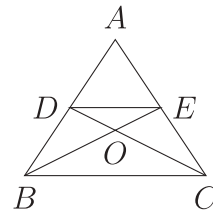
3. Prove that if each of the two triangles is similar to a third triangle, they are congruent to each other.
4. Prove that, if one acute angle of a right-angled triangle is equal to an acute angle of another right-angled triangle, the triangles are similar.
5. Prove that the two right-angled triangles formed by the perpendicular from the vertex containing the right angle are similar to each other and also to the original triangle.

6. In the adjacent figure, $\angle B = \angle D$ and $CD = 4AB$. Prove that, $BD = 5BL$.

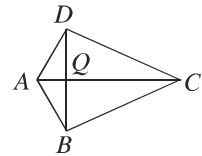


7. A line segment drawn through the vertex A of the parallelogram $ABCD$ intersects the BC and DC at M and N respectively. Prove that $BM \times DN$ is a constant.
8. If the heights of the equiangular triangles ABC and DEF are AM and DN respectively, prove that $AM : DN = AB : DE$.

9. In the adjacent figure $BC \parallel DE$
- 1) Prove that, $\triangle BOC$ and $\triangle DOE$ are similar.
 - 2) Prove that, $AD : BD = AE : CE$.
 - 3) Prove that, $BO : OE = CO : OD$

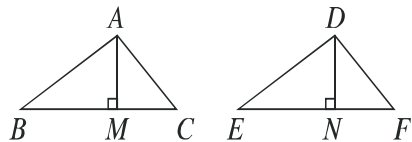


10. In the adjacent figure $BD \perp AC$ and $DQ = BQ = 2AQ = \frac{1}{2}QC$. Prove that, $DA \perp DC$.



11. In the triangles $\triangle ABC$ and $\triangle DEF$, $\angle A = \angle D$. Prove that, $\triangle ABC : \triangle DEF = AB \cdot AC : DE \cdot DF$
12. In the figure, ABC and DEF are two similar triangles.

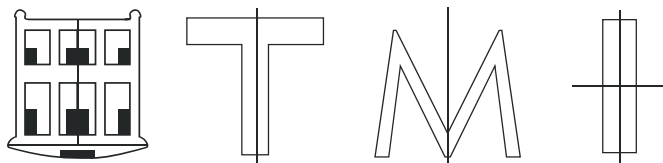
- 1) Name the matching sides and matching angles of the triangles.
- 2) Prove that,
$$\frac{\triangle ABC}{\triangle DEF} = \frac{AB^2}{DE^2} = \frac{AC^2}{DF^2} = \frac{BC^2}{EF^2}$$



- 3) If $BC = 3$ c m, $EF = 8$ c m, $\angle B = 60^\circ$, $\frac{BC}{AB} = \frac{3}{2}$ and area of $\triangle ABC$ is 3 square c m, then draw the triangle $\triangle DEF$ and find its area.

Symmetry

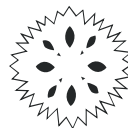
Symmetry is an important geometrical concept, commonly exhibited in nature and is used almost in every field of our activity. Artists, designers, architects, carpenters always make use of the idea of symmetry. Tree-leaves, flowers, beehives, houses, tables, chairs -everywhere we find symmetrical designs. A figure has line symmetry, if there is a line about which the figure may be folded so that the two parts of the figure will coincide.



Each of the above figures has the line of symmetry.

Work:

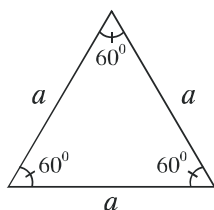
- 1) Sumi has made some paper-cut design as shown in the adjacent figure. In the figure, mark the lines of symmetry. How many lines of symmetry does the figure have?
- 2) Write and identify the letters in English alphabet having line symmetry. Also mark their line of symmetry.



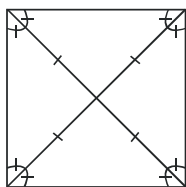
Line of symmetry of a regular polygon

A polygon is a closed figure made of several line segments. A polygon is said to be regular if all its sides are of equal length and all its angles are equal. The triangle is a polygon made up of the least number of line segments. An equilateral triangle is a regular polygon of three sides. An equilateral triangle is regular because its sides as well as angles are equal. A square is the regular polygon of four sides.

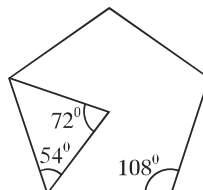
The sides of a square are equal and each of the angles is equal to one right angle. Similarly, in regular pentagons and hexagons, the sides are equal and the angles are equal as well.



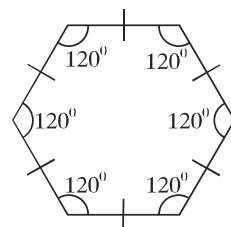
Equilateral triangle



Square



Regular pentagon



Regular hexagon

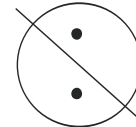
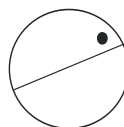
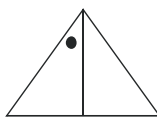
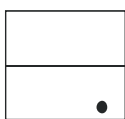
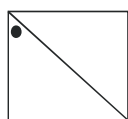
Each regular polygon is a figure of symmetry. Therefore, it is necessary to know their lines of symmetry. Each regular polygon has many lines of symmetry as it has many sides.

Three lines of symmetry	Four lines of symmetry	Five lines of symmetry	Six lines of symmetry
Equilateral triangle	Square	Regular pentagon	Regular hexagon

The concept of line symmetry is closely related to mirror reflection. A geometrical figure has line symmetry when one half of it is the mirror image of the other half. So, the line of symmetry is also called the reflection symmetry.

**Work:**

- 1) The line of symmetry is given, find the other half.



- 2) Identify the lines of symmetry in the following geometrical figures.

(1) An isosceles triangle

(2) A scalene triangle

(3) A square

(4) A rhombus

(5) A regular hexagon

(6) A pentagon

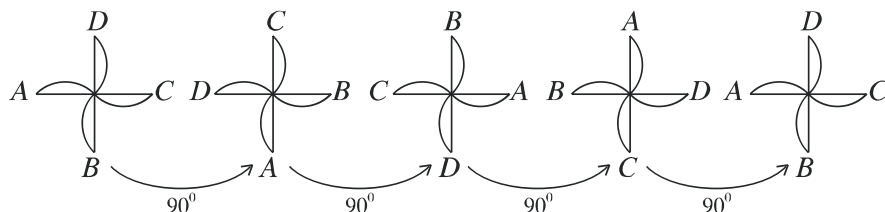
(7) A circle

Rotational symmetry

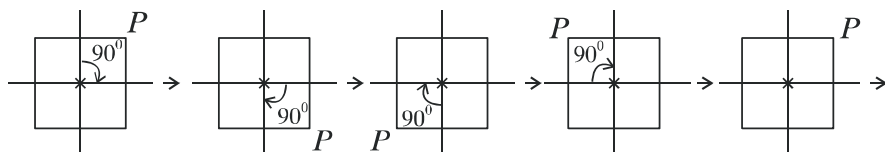
When an object rotates around any fixed point, its shape and size do not change. But the different parts of the object change their position. If the new position of the object after rotation becomes identical to the original position, we say the object has a rotational symmetry. The wheels of a bicycle, ceiling fan, square, etc. are examples of objects having rotational symmetry. As a result of rotation the blades of the fan looks exactly the same as the original position more than once. The blades of a fan may rotate in the clockwise direction or in the anticlockwise direction. The wheels of a bicycle may rotate in the clockwise direction or in the anticlockwise direction. The rotation in the anticlockwise direction is considered the positive direction of rotation.

This fixed point around which the object rotates is the **centre of rotation**. The angle of turning during rotation is called the angle of rotation. A full-turn means rotation by 360° ; a half-turn is rotation by 180° .

In the figure below, a fan with four blades rotating by 90° is shown in different positions. It is noted during a complete revolution in four positions (rotating about the angle by 90° , 180° , 270° and 360°), the fan looks exactly the same. For this reason, it is said that the rotational symmetry of the fan is order 4.



Here is one more example for rotational symmetry. Consider the intersection of two diagonals of a square the centre of rotation. In the quarter turn about the centre of the square, any diagonal position will be as like as the second figure. In this way, when you complete four quarter-turns, the square reaches its original position. It is said that a square has a rotational symmetry of order 4.

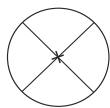


Observe also that every object occupies same position after one complete revolution. So every geometrical object has a rotational symmetry of order 1. For finding the rotational symmetry of an object, one needs to consider the following matter:

- 1) The centre of rotation
- 2) The angle of rotation
- 3) The direction of rotation
- 4) The order of rotational symmetry

Work:

- 1) Give examples of 5 plane objects from your surroundings which have rotational symmetry.
- 2) Find the order of rotational symmetry of the following figures.



(a)



(b)



(c)



(d)



(e)

Line symmetry and rotational symmetry

We have seen that some geometrical shapes have only line symmetry, some have only rotational symmetry and some have both line symmetry and rotational symmetry. For example, the square has four lines of symmetry as well rotational symmetry of order 4.

The circle is the most symmetrical figure, because it can be rotated around its centre through any angle. Therefore, it has unlimited order of rotational of symmetry. At the same time, every line through the centre forms a line of reflection symmetry and so it has unlimited number of lines of symmetry.

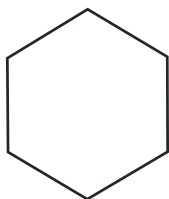
Work: Determine the line of symmetry and the rotational symmetry of the given alphabet and complete the table below: (one is done for you)

Letter	Line of symmetry	Number of lines of symmetry	Rotational symmetry	Order of rotational symmetry
Z	NO	0	YES	2
H				
O				
E				
C				

Exercise 14.3

- How many lines of symmetry does a scalene triangle have?
 1) 0 2) 1 3) 3 4) Countless

From the following figure, answer question 3 and 4.



The length of each side of the polygon is 6 c.m.

- How many lines of symmetry does the polygon have?
 1) 3 2) 6 3) 7 4) Countless
- For the polygon-
 (i) Order of rotational symmetry is 4
 (ii) The angle of rotation is 60°
 (iii) Each angles in equal

Which of the following is true?

- 1) *i* 2) *ii* 3) *ii* and *iii* 4) *i*, *ii* and *iii*

11. When you slice a lemon, the cross-section looks as shown in the figure. Determine the rotational symmetry of the figure.



12. Fill in the blanks:

Shape	Centre of Rotation	Order of Rotation	Angle of rotation
Square			
Rectangle			
Rhombus			
Equilateral triangle			
Semi-circle			
Regular pentagon			

13. Name the quadrilaterals which have line of symmetry and rotational symmetry of order more than 1.
14. Can we have a rotational symmetry of a body of order more than 1 whose angle of rotation is 18° ? Justify your answer.

Sample Questions

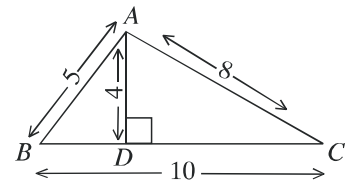
Multiple Choice Questions:

1. If triangles $\triangle ABC$ and $\triangle PQR$ have proportional areas and heights, and the height of $\triangle ABC$ is a units, what is the height of $\triangle PQR$?
 (a) a (b) $2a$ (c) $3a$ (d) $4a$
2. In plane geometry:
 (i) The triangle is a polygon made up of the least number of line segments.
 (ii) A rhombus is the regular polygon of four sides
 (iii) Sides of a regular polygon are equal, but angles are not

Which of the following is true?

- (a) i (b) i and ii (c) i and iii (d) i , ii and iii

Use the information from the adjacent figure to answer the questions 3 and 4:



3. What is the ratio of the height and base of $\triangle ABC$?

- 1) $\frac{1}{2}$ 2) $\frac{4}{5}$ 3) $\frac{2}{5}$ 4) $\frac{5}{4}$

4. What is the area of $\triangle ABD$ in square units?

- 1) 6 2) 20 3) 40 4) 50

Creative Question:

5. The angle bisector of $\angle A$ in triangle $\triangle ABC$, namely AD , intersects BC at point D . A line segment CE , parallel to DA , intersects the extended side BA at point E .

- In triangles $\triangle PQR$ and $\triangle LMN$, the bases are QR and MN respectively, and $QR = MN$. Show that the ratio of their areas equals the ratio of their heights.
- Prove that $BD : DC = BA : AC$.
- If a line parallel to BC intersects AB and AC at P and Q respectively, prove that $BD : DC = BP : CQ$.

Short-Answer Questions:

- In triangle $\triangle ABC$, E is drawn parallel to BC and intersects AB and AC at D and E respectively and if $AB : AD = 2 : 1$ show that, $DE : BC = 1 : 2$
 - If $\triangle PQR$ and $\triangle LMN$ have equal heights, show that the ratio of their areas equals the ratio of their bases.
 - In trapezium $ABCD$ where $AB \parallel DC$ and diagonals AC and BD intersect at O , show that triangles $\triangle OAB$ and $\triangle OCD$ are similar.
 - If triangles ABC and DEF are similar and a pair of corresponding sides AB and DE are in the ratio $3:4$, find the ratio of the areas $\triangle ABC : \triangle DEF$.